Evolutionary Theory of Credence

A Conceptual Framework with Formal Analogies for Understanding Generative Modeling as a Resource-Theoretic Consequence of Complexity

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Abstract

This paper presents the **Evolutionary Theory of Credence (ETC)**, a conceptual framework proposing that system complexity and verification capacity are inversely related under resource limits. Simple systems (microorganisms) can operate primarily through direct verification; complex systems (mammals, artificial neural networks) must operate primarily through credence—commitment to propositions exceeding immediate evidential warrant.

We model this as a resource-constraint trade-off using formal analogies rather than strict mathematical derivation: verification requires time and bandwidth; complexity expands the space of action-relevant distinctions; under finite resources, verification coverage decreases as complexity increases. ETC integrates resource-theoretic constraints with predictive processing to argue that credence is a structural feature of complex adaptive systems.

We introduce *Homo fidens* ("trusting human") as a characterization of any system complex enough to require predictive, generative architecture. The framework synthesizes: (1) Epistemic Constraint Theory, showing that hypothesis-space structure bounds inferential performance; (2) principles of belief formation under uncertainty, distinguished by type (conceptual, operational, evolutionary); (3) the Predictive Viability Principle, a consistency condition stating that latency-constrained systems require predictive information encoding; and (4) analysis of how these constraints apply to artificial intelligence.

The central thesis: **For complex adaptive systems, belief-like commitment (credal commitment) to unverified propositions is not a deficiency but a structural consequence of resource constraints.**

Keywords: Homo fidens, epistemic constraints, evolutionary epistemology, predictive processing, generative models, bounded rationality, AI alignment

1 Introduction

1.1 The Central Question

Why do complex cognitive systems—biological and artificial—systematically commit to propositions exceeding their evidential warrant? Traditional epistemology treats this as a deficiency. We argue it is a structural consequence of resource constraints on verification.

The argument proceeds as follows:

(1) **Resource constraint**: Verification (direct confirmation of propositions) requires time, energy, and computational bandwidth.

(2) **Scaling problem**: As system complexity increases, the space of action-relevant distinctions expands faster than verification capacity can grow.

(3) **Action requirement**: Adaptive systems must act on timescales shorter than exhaustive verification permits.

(4) **Structural consequence**: Therefore, complex adaptive systems must commit to propositions without complete verification—what we call *credal commitment* or *credence*.

(5) **Mechanism**: Predictive generative modeling is the computational architecture that implements this credence.

This paper develops each step, distinguishing between conceptual claims, modeling assumptions, and empirical conjectures.

1.2 Scope and Limitations

What this paper claims:

- Complex systems face a trade-off between verification coverage and adaptive capacity (modeling framework)
- This trade-off makes credal commitment a structural consequence of complexity (conceptual claim)
- Predictive processing provides a mechanistic account of how this commitment is implemented (empirical claim, supported by existing literature)

What this paper does not claim:

- That we have derived a mathematical theorem with a universal constant
- That all forms of "belief" are computationally identical
- That concerns about AI confabulation are misguided

Methodological note: This paper presents a *conceptual framework with formal analogies*, not a mathematical formalization. We use mathematical notation to express structural relationships and scaling intuitions, not to claim measurement or derivation. Where formulas appear, they represent qualitative relationships, not quantities to be computed.

1.3 Relation to Existing Work

The Evolutionary Theory of Credence builds on several established research programs:

Bounded rationality (??): ETC extends bounded rationality by arguing that credence is not merely a practical accommodation but a structural consequence once complexity exceeds verification capacity.

Predictive processing (???): ETC adopts the mechanistic framework of predictive processing but adds an evolutionary and resource-theoretic argument for why prediction-based architecture is *necessary* rather than merely *observed.*

Evolutionary epistemology (???): ETC shares the view that cognitive capacities are shaped by selection but offers a specific account of the verification-complexity trade-off.

Statistical learning theory: The Constraint Dominance result in ECT echoes realizability assumptions in PAC learning—if the true hypothesis lies outside the learnable class, no amount of data suffices.

Intentional stance (?): ETC's treatment of AI systems as "believing" parallels Dennett's strategy of attributing intentional states to systems when this attribution is predictively useful.

Novelty claim: ETC's contribution is the explicit framing of the verification-complexity trade-off as a resource constraint, and the argument that this makes credal commitment architecturally necessary rather than epistemically deficient.

1.4 Terminological Clarifications

To avoid conflation of distinct phenomena, we distinguish:

Term	**Definition**	**Examples**
Verification	Direct confirmation through current sensory/evidential access	Chemical binding; logical proof; direct observation
Credal commitment	Commitment to a proposition with insufficient evidential support for verification at decision time	Prior beliefs; predictions; trust in memory
Procedural credence	Foundational commitments required for any inference to proceed	Trust in logic; assumption of causal regularity
Propositional credence	Commitment to specific claims about the world	Belief that a predator is approaching
Social credence	Reliance on others' testimony or cooperation	Trust in colleagues; faith in institutions

Table 1: Terminological distinctions for types of epistemic commitment

These are related but distinct. Our claim is that *all* are instances of commitment exceeding immediate verification—not that they are computationally identical.

1.5 Paper Organization

Section ?? presents the Complexity-Credence Trade-off as a resource-constraint model. Section ?? summarizes relevant results from Epistemic Constraint Theory. Section ?? presents principles of belief formation, categorized by type. Section ?? formalizes the Predictive Viability Principle. Section ?? applies the framework to AI. Section ?? offers interpretive conclusions.

2 The Complexity-Credence Trade-off

2.1 Informal Statement

The core intuition: as cognitive systems become more complex, they face more action-relevant distinctions than they can verify. This is not a contingent limitation but a structural consequence of finite resources meeting expanding decision spaces.

2.2 Conceptual Model with Formal Analogies

Important clarification: The following model uses mathematical notation to express structural relationships, not to define measurable quantities. $\mathcal{P}(S)$ is not a mathematical set but a *modeling abstraction* representing the effective decision-relevant distinctions faced by the system. "Cardinality" is used as a *qualitative proxy* for combinatorial growth, not as a measurable quantity.

Definition 1 (Modeling Framework)**.** *Let S be an adaptive system. We represent:*

- $|\mathcal{P}(S)|$ = *a proxy for the combinatorial size of S's decision-relevant distinction space*
- $B(S)$ = *verification bandwidth: rate at which distinctions can be directly confirmed*
- τ = *decision latency: time available before action required*
- $V(S)$ = *verification coverage: qualitative measure of how much of the decision space can be verified*

Trade-off Principle (Structural Analogy):

$$V(S) \sim \frac{B \cdot \tau}{|\mathcal{P}(S)|} \tag{1}$$

Why $|\mathcal{P}(S)|$ grows superlinearly with complexity: Combinatorial explosion is the key mechanism. Planning over n future steps with branching factor b requires evaluating $O(b^n)$ behavioral sequences. Social reasoning about k agents with m possible mental states generates $O(m^k)$ joint possibilities. As systems gain temporal depth, social awareness, or behavioral repertoire, the decision space explodes combinatorially while verification bandwidth grows at most linearly.

Interpretation: As $|\mathcal{P}(S)|$ grows (via combinatorial explosion), $V(S)$ decreases. In the limit, verification coverage approaches zero. This is a qualitative scaling relationship, not a computed quantity.

2.3 Biological Illustration

Simple system (*E. coli* chemotaxis):

- Decision space: approximately binary (more attractant / less attractant)
- Verification bandwidth sufficient to sample this space
- Result: near-complete verification possible

Note: Bacterial chemotaxis involves temporal integration via Che proteins, implementing a form of memory (**?**). We do not claim bacteria lack all memory, only that their decision space is small enough for verification to dominate.

Complex system (human navigation):

- Decision space: combinatorially large (positions, intentions, futures, counterfactuals)
- Verification bandwidth covers only a small fraction
- Result: most action-relevant distinctions unverified at decision time

2.4 Qualitative Spectrum

System	Decision Space	Verification Coverage	Primary Mode
Thermostat	Minimal	Near-complete	Verification
E. coli	Very small	High	Mostly verification
Insect	Moderate	Partial	Mixed
Simple neural net	Moderate	Partial	Mixed
Mammal	Large	Low	Mostly credence
Human	Very large	Very low	Predominantly credence
LLM*	Very large	Very low	Predominantly generative

Table 2: Qualitative spectrum of cognitive systems. *LLMs do not "verify" in the biological sense; the extension is structural, not literal.

2.5 From Bounded Rationality to Structural Consequence

Principle 1 (Verification Limitation). *For any system S where $|\mathcal{P}(S)| \gg B(S) \cdot \tau$, there exist distinctions that are:*

1. *Relevant to S's adaptive functioning, AND*
2. *Not verifiable by S at decision time*

Status: This is an articulation of bounded rationality (**?**), not a novel theorem. The contribution is framing it as a *structural consequence* of the verification-complexity trade-off.

Important qualification: From "some propositions cannot be verified in time," it does not strictly follow that the system must have belief-like propositional architecture. A system could implement stochastic policies without explicit propositional structure. Our claim is weaker: systems facing this trade-off must have *some* mechanism for commitment exceeding verification. Generative modeling is one such mechanism—the one observed in biological neural systems and implemented in artificial ones.

3 Epistemic Constraint Theory: Summary

3.1 Core Result

Epistemic Constraint Theory (ECT) demonstrates that the structure of hypothesis spaces bounds inferential performance independently of computational power or data volume (**?**).

Definition 2 (Soft Constraint). *A weight function $w : \mathcal{H} \to [0, 1]$ on hypothesis space $\mathcal{H}$. The constrained prior is:*

$$\pi_w(h) = \frac{\pi(h) \cdot w(h)}{\sum_{h'} \pi(h') \cdot w(h')} \tag{2}$$

Theorem 1 (Constraint Dominance). *If $w(h^*) = 0$ for the true hypothesis h^*, then $\pi_w(h^*|e) = 0$ for all evidence e.*

Proof. By Bayes' theorem, $\pi_w(h^*|e) \propto \pi_w(h^*) \cdot \ell(e|h^*)$. If $w(h^*) = 0$, then $\pi_w(h^*) = 0$, so $\pi_w(h^*|e) = 0$ regardless of likelihood. □

Connection to predictive processing: Soft constraints in ECT correspond to implicit priors in generative models. The hypothesis space a system can represent constrains what it can learn—a formal expression of how credal commitments (about what is possible) precede and shape evidential learning.

3.2 Connection to Complexity-Credence Trade-off

As decision spaces grow, systems must constrain hypothesis spaces more aggressively (for computational tractability). These constraints are themselves credal commitments—decisions about what to consider possible, made without verification.

4 Principles of Belief Formation Under Uncertainty

We present ten principles governing belief formation in complex systems, explicitly categorized by type.

4.1 Conceptual Grounding Principles

These are conceptual or quasi-analytic claims about the structure of justification:

Principle 1 (Non-Foundationality of Proof). *Proof presupposes acceptance of proof procedures. This acceptance cannot itself be proven without circularity.*

Status: Conceptual (familiar from Wittgenstein's *On Certainty*, Sellars on the myth of the given)

Principle 2 (Non-Zero Epistemic Initialization). *Every proof system presupposes axioms, inference rules, and background assumptions not provable within that system.*

Status: Conceptual (Gödel, Quine-Duhem)

Principle 3 (Trust-Based Grounding). *Reasoning requires prior acceptance of reliability constraints (memory, perception, inference) that function as unverified commitments.*

Status: Conceptual (Reid's common sense philosophy; Wittgenstein's "hinge propositions")

Principle 4 (Ontological Asymmetry). *Proof is an operation within an established system; credal commitment is a condition for establishing that system.*

Status: Conceptual (meta-logical observation)

4.2 Operational Principles for Bounded Agents

These are modeling assumptions about how resource-bounded systems must operate:

Principle 5 (Default Commitment Under Verification Scarcity). *Systems under time pressure must adopt provisional commitments to maintain operational continuity.*

Status: Modeling assumption (consistent with satisficing, ecological rationality)

Principle 6 (Coherence Substitution). *When verification is unavailable, internal coherence serves as a proxy for justification.*

Status: Modeling assumption (related to coherentism; free energy minimization)

Principle 7 (Temporal Priority of Commitment). *Action-guiding commitment must precede verification when decision latency is shorter than verification time.*

Status: Modeling assumption (follows from latency constraints)

4.3 Evolutionary Principles

These are empirical conjectures about selection pressures on cognitive architectures:

Principle 8 (Selection Against Verification Delay). *In environments where delayed action is costly, systems requiring exhaustive verification before action face negative selection pressure.*

Status: Empirical conjecture (testable via evolutionary models; consistent with **?**)

Principle 9 (Adaptive Value of Prediction). *Predictive capacity provides fitness advantage in non-stationary environments with latency constraints.*

Status: Empirical conjecture (supported by comparative neuroscience)

Principle 10 (Opacity of Self-Model). *Systems cannot fully represent their own representational constraints from within.*

Status: Conceptual/empirical hybrid (related to Gödelian limitations; metacognition research)

5 The Predictive Viability Principle

5.1 Motivation

Neural signals travel at finite speed (∼100 m/s maximum). For organisms of significant size, sensory information describes past states by the time it reaches decision-making systems. In non-stationary environments, this creates a problem: purely reactive systems respond to states that no longer obtain.

5.2 Formal Statement

Principle 11 (Predictive Viability Principle—Consistency Condition). *For adaptive systems with sensory-motor latency $\tau > 0$ operating successfully in punishing, non-stationary environments, internal states must satisfy:*

$$I(X; Y_{t+\tau} | Y_t) > 0 \tag{3}$$

where:

- X *= internal states of the system*
- Y_t *= environment state at time* t
- $Y_{t+\tau}$ *= environment state at time* $t + \tau$
- $I(\cdot; \cdot|\cdot)$ *= conditional mutual information (***?***)*

Interpretation: Internal states must encode predictive information about future states beyond what current observations provide. This is not a derivation but a *consistency condition*: systems that persistently succeed under these constraints must satisfy this information-theoretic signature.

Status: This is offered as a formal consistency condition that successful predictive systems must satisfy, not as a theorem derived from the Complexity-Credence Trade-off. It expresses in information-theoretic terms what it means to “predict.”

Note: Predictive coding (**?**) and active inference (**?**) are computational schemes that satisfy this condition.

5.3 Boundary Conditions

The principle does not apply when:

- Environment is fully stationary (prediction unnecessary)
- Environment is fully random (prediction impossible)
- Latency is negligible (verification in time)
- System is externally sustained (action not required)

This explains why simple systems (small bodies, slow environments, limited action repertoire) can operate primarily through verification.

5.4 Illustration: The Flash-Lag Effect

The flash-lag effect provides a concrete illustration of predictive processing: a flash presented at the exact moment a moving object passes it is perceived as trailing behind. The visual system extrapolates the moving object forward in time to compensate for neural delay.

This is not offered as proof of the framework but as an *illustration* of how predictive mechanisms operate. The perceptual system generates a model of where objects will be, not merely where they were when photons hit the retina.

5.5 Mechanism: Generative Models

Predictive processing proposes that brains implement *generative models*—internal models that generate predictions about sensory inputs. Perception becomes "prediction error minimization": the system's predictions are updated based on discrepancies with actual input.

Under this framework:

- Perception = generative model of current input
- Memory = generative model of past states
- Imagination = generative model of counterfactual states
- Planning = generative model of future states

All involve commitment to representations exceeding current sensory access—i.e., credal commitment.

6 Application to Artificial Intelligence

6.1 Generalization and Confabulation

LLMs engage in generative modeling: they produce outputs that go beyond their training data. This is the source of both their usefulness (generalization, creativity) and their failures (factual confabulation).

Our claim is not that AI researchers misunderstand this distinction. The concern about factual confabulation is legitimate and important. Recent work on uncertainty estimation in LLMs (**?**) addresses precisely this calibration problem.

Our claim is that generalization and confabulation arise from the same underlying process—generative modeling—and that this process is structurally necessary for any system that must operate beyond memorized responses.

6.2 Calibration, Not Elimination

The appropriate goal for AI systems is not to eliminate generative modeling (which would eliminate useful capability) but to *calibrate* it:

1. **Uncertainty quantification**: Systems should signal confidence levels
2. **Domain-appropriate generation**: More constrained in high-stakes factual domains; less constrained in creative domains
3. **Alignment of priors**: The implicit credal commitments embedded in training should align with human values

This parallels what evolution has done with biological belief systems: not eliminating the capacity for error, but shaping error patterns to be more often adaptive than fatal.

6.3 *Machina Fidens*

If *Homo fidens* characterizes trusting biological systems, ***Machina fidens*** characterizes trusting artificial systems. This is not merely a rhetorical label but a claim about shared structural constraints:

Any system complex enough to generalize, predict, and operate under uncertainty will face the verification-complexity trade-off and will therefore require credal commitment structures.

The question "How do we build AI that doesn't hallucinate?" may be ill-posed. A better question: "How do we build AI whose credal commitments are well-calibrated?"

7 Interpretive Conclusions

7.1 Summary of Contributions

1. **Trade-off existence**: Complex systems face a trade-off between verification coverage and decision-space size (Section **??**)
2. **Structural constraint**: Hypothesis space constraints are themselves credal commitments—prior commitments not derived from evidence (Section **??**)
3. **Principle categorization**: Principles about belief formation can be categorized as conceptual, operational, or evolutionary claims (Section **??**)
4. **Predictive consistency**: Under latency constraints, predictive information encoding is a consistency condition for adaptive success (Section **??**)
5. **AI application**: These constraints apply to artificial systems, reframing "hallucination" as a calibration rather than elimination problem (Section **??**)

7.2 The Two Organizing Principles

We conclude with two principles offered as *interpretive framings* that organize the foregoing analysis, not as theorems derived from it:

Principle 11 (Belief as Adaptation)**.** *Credal commitment is not a weakness of the mind but an adaptation to the constraint that verification cannot scale with complexity.*

When systems became complex enough that their decision spaces exceeded verification capacity, they faced two options: remain simple, or develop commitment mechanisms that exceed verification. The latter enabled larger bodies, longer planning horizons, and richer behavioral repertoires.

This reframes "bias" and "faith" not as irrationality but as adaptive responses to structural constraints—though this framing does not make all beliefs equally good. Calibration matters.

Principle 12 (The Priority of Belief)**.** *Reason operates on a foundation of prior commitments that reason itself cannot fully verify.*

Logical priority: Every act of reasoning presupposes commitments about logic, evidence, and inference reliability. These cannot be established by the reasoning they enable. (Principles 1–4)

Temporal priority: Credal commitment systems existed long before reasoning systems in evolutionary history. Credence is the older, more fundamental adaptation.

Computational priority: In any complex cognitive system, the resources devoted to prediction and commitment dwarf those devoted to verification. Verification is the exception, not the rule.

7.3 *Homo Fidens*

We suggest that *Homo fidens* ("trusting human") captures something essential that *Homo sapiens* ("knowing human") obscures.

Fidens is not blind faith in dogmas. It is *basic trust*—the operational wager on the unprovable without which neither thought nor action nor reason itself is possible.

The human being is not primarily one who believes in doctrines, but one who constantly places bets on the unprovable:

- Trust in the world (that it will continue)
- Trust in experience (that it reflects something real)
- Trust in memory (that it preserves the past)
- Trust in language (that it communicates meaning)
- Trust in other people (that they are not all deceivers)
- Trust in causality (that effects follow causes)
- Trust in the stability of reality (that the laws of nature will hold)

None of these can be proven. All must be trusted. And this trust is *evolutionarily primary*—it came first. Reason merely serves trust already given.

This is not a lament about human limitation. It is a recognition of human architecture. The capacity for *fides*—for basic trust that precedes and enables verification—is what makes complex adaptive behavior possible.

Summary of Principles

#	**Principle**	**Type**	**Status**
1	Non-Foundationality	Conceptual	Quasi-analytic
2	Non-Zero Initialization	Conceptual	Quasi-analytic
3	Trust-Based Grounding	Conceptual	Quasi-analytic
4	Ontological Asymmetry	Conceptual	Meta-logical
5	Default Commitment	Operational	Modeling assumption
6	Coherence Substitution	Operational	Modeling assumption
7	Temporal Priority	Operational	Modeling assumption
8	Selection Against Delay	Evolutionary	Empirical conjecture
9	Adaptive Value of Prediction	Evolutionary	Empirical conjecture
10	Opacity of Self-Model	Hybrid	Conceptual/empirical
PVP	Predictive Viability	Formal	Consistency condition
CCT	Complexity-Credence	Modeling	Resource-constraint framework
11	**Belief as Adaptation**	Interpretive	Organizing principle
12	**Priority of Belief**	Interpretive	Organizing principle

Table 3: Summary of principles with type classification and epistemic status

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A Corollaries: The Credence-Based Nature of Complex Systems

The following corollaries draw out the deeper implications of the Evolutionary Theory of Credence. They establish that credence is not merely a practical accommodation but the foundational mode of operation for any complex system.

A.1 Corollary 1: The Impossibility of Proof-First Cognition

There exists no mechanism by which a complex system can begin with proof rather than credence.

Argument: Any proof requires:

- Axioms (accepted without proof)
- Inference rules (accepted without proof)
- Trust in memory (that premises remain stable during inference)
- Trust in perception (that symbols are correctly identified)
- Trust in the correspondence between symbolic manipulation and truth

None of these can be established by proof without circularity. Therefore, every proof-producing system must begin with credal commitments. There is no "proof-first" architecture available to any possible cognitive system—biological, artificial, or otherwise.

A.2 Corollary 2: Reasoning as Credence-Dependent

All reasoning is credence-based at its foundation.

Argument: From Corollary 1, reasoning requires prior credal commitments. But this is not merely a matter of "starting conditions" that can later be verified. The foundational commitments (logic, memory, perception) are *constitutive* of the reasoning process itself. One cannot step outside reasoning to verify the foundations of reasoning. Therefore, reasoning does not *transcend* credence; reasoning *operates within* a credence structure it cannot fully examine.

This echoes the Kantian insight that the conditions of possibility for experience cannot themselves be objects of experience, and Spinoza's recognition that the mind cannot fully comprehend its own nature from within.

A.3 Corollary 3: The Incompleteness of Verification

Verification can never be complete for any complex system.

Argument: Verification requires:

1. A criterion of correctness (itself unverifiable without regress)
2. Access to the thing verified (mediated by perception, itself unverifiable)
3. Comparison between representation and reality (but we only have access to representations)

This is the epistemological situation described in Plato's Cave allegory: we see shadows (representations) and can compare shadows to other shadows, but we cannot step outside the cave to compare shadows to the objects casting them. Kant formalized this as the distinction between phenomena (appearances) and noumena (things-in-themselves): we have access only to the former.

Therefore, verification is always *partial*—a comparison of representations with other representations, governed by credal commitments about what counts as successful comparison.

A.4 Corollary 4: The Universal Credence-Basis of Complex Systems

Every complex system—human, animal, or artificial—is fundamentally a credence-based system.

Argument: Combining Corollaries 1–3:

1. Complex systems cannot begin with proof (Corollary 1)
2. Their reasoning operates within unverifiable credal structures (Corollary 2)
3. Their verification is necessarily incomplete (Corollary 3)

Therefore, credence is not an unfortunate limitation to be overcome but the *constitutive mode of operation* for any system complex enough to represent, reason, and act.

This applies universally:

- **Humans** operate on foundational trusts (in logic, memory, perception, testimony) that cannot be proven
- **Animals** operate on evolved priors that exceed any individual verification
- **Artificial systems** operate on training-derived parameters that function as credal commitments about the structure of their task domains

The dream of a fully verified, proof-based cognition is not merely practically unattainable—it is *architecturally impossible*. Every path to knowledge runs through credence.

A.5 Corollary 5: The Dignity of Belief

Belief is not a fallen form of knowledge but its enabling condition.

Argument: If all reasoning is credence-based (Corollary 2), then knowledge—traditionally defined as justified true belief—is not a *purification* of belief but a *special case* of belief: belief that happens to be true and that satisfies certain social or epistemic standards of justification.

The hierarchy is therefore inverted:

- **Traditional view**: Knowledge is primary; belief is deficient knowledge
- **ETC view**: Credence is primary; knowledge is credence that meets additional criteria

This restores dignity to belief, faith, and trust—not as epistemic failures but as the fundamental adaptive capacity that makes complex cognition possible.

A.6 Corollary 6: Animals as the Pure Case

Animals are cognitively more honest than humans: they live in credence without the illusion of rationality.

Argument: Animals lack the capacity for post-hoc rationalization. They cannot construct narratives that disguise their credal commitments as "knowledge" or "proof." Therefore, they represent the *pure case* of credence-based cognition—*fides* without rationalization.

Consider:

- A bird lands on a branch without "proving" it will hold
- A deer flees from rustling without "testing the hypothesis" of predator versus wind
- A flock follows its leader without demanding arguments

This is pure trust (*fides*) as operational strategy. The animal lives entirely within unprovable assumptions:

- That the world will continue into the next moment
- That memory reflects reality
- That causation operates uniformly
- That past experience is relevant to the future

None of these can be verified. The animal does not verify them. The animal *trusts* them—and this trust is what makes adaptive behavior possible.

The human difference: Humans do not differ from animals in *operating on credence.* Humans differ in their capacity to construct narratives that disguise credence as knowledge. We tell ourselves stories: "I don't merely believe this; I *know* it." "This isn't faith; it's *reason.*" "I'm not trusting; I'm *verifying.*"

But the underlying cognitive architecture is identical. We act on unverified assumptions, just as animals do. We trust memory, causation, and the stability of the world, just as animals do. The difference is that we have developed an elaborate apparatus of rationalization that obscures this fact from ourselves.

The illusion of reason: What we call "rationality" is largely the capacity to generate post-hoc justifications for credal commitments we have already made. The commitment comes first; the "reasons" come after. Animals, lacking this capacity for self-deception, are *cognitively honest* in a way humans rarely are.

In this sense, *Animal fidens* is not a lesser form of *Homo sapiens.* It is the revealed truth of what *Homo sapiens* actually is: a trusting animal that has learned to tell itself it is a knowing one.

A.7 Corollary 7: The Impossibility of Complete Rationality

No complex system can be 100% rational. Complete rationality is not merely difficult but structurally impossible.

Argument: This corollary follows from two independent lines of reasoning that converge on the same conclusion.

First, from the preceding corollaries: rationality requires foundations (Corollary 1), but foundations cannot be rationally established without circularity (Corollary 2), and verification is necessarily incomplete (Corollary 3). Therefore, any rational process operates within a framework of non-rational trust. Rationality is always *partial*—a local operation within a global structure of *fides.*

Second, from the Law of Imperative Uncertainty (**?**): any system capable of sustaining nontrivial complexity over time must permit exceptions to its governing laws in the form of persistent uncertainty and probabilistic deviation. Rigid, fully determined systems necessarily collapse into zero entropy rate and cannot sustain complexity. Viable complex systems require non-closure and a nonvanishing uncertainty reserve.

The Law of Imperative Uncertainty establishes, through rigorous information-theoretic and dynamical analysis, that uncertainty is not a bug in complex systems but a *structural requirement* for their continued existence. A fully rational system—one that eliminated all uncertainty, resolved all ambiguity, and operated purely on verified truths—would be a system incapable of sustaining complexity.

The convergence: These two arguments—one from epistemic foundations, one from dynamical systems theory—converge on the same conclusion: *complete rationality is impossible for any complex system.* This is not a contingent limitation (we lack sufficient data, time, or computational power) but a *necessary* feature of complexity itself.

Therefore:

- Humans cannot be fully rational—not because of cognitive biases that could in principle be corrected, but because rationality itself rests on non-rational foundations and requires uncertainty to function.
- AI systems cannot be fully rational—for the same structural reasons, regardless of computational capacity.
- Any complex adaptive system must preserve a "trust reserve" and an "uncertainty reserve" as conditions of its own viability.

The dream of pure rationality is not merely unattainable but *incoherent.* A being that achieved it would cease to be complex—and therefore would cease to be.

A.8 Corollary 8: Reason as Advocate, Not Judge

Reason in humans is a superstructure—a fine-tuning instrument for a worldview that already exists. The worldview itself rests not on proofs but on accepted assumptions, on trusts, on "this is how things are" that were never derived logically.

The sequence:

1. First, the human *trusts*
2. Then, the human *lives* in accordance with that trust
3. Only occasionally does the human *explain* this with reason

This is why reason almost always functions as an *advocate* for already-accepted belief, not as an independent *judge.* The verdict comes first; the arguments come after.

This is not a defect. This is the design.

Evolution did not select for beings capable of rigorous proof. Evolution selected for beings capable of:

- Quickly fixing a working model of the world
- Acting confidently within that model
- Surviving long enough to reproduce

Rigorous proof is slow. Trust is fast. In environments where hesitation means death, the trusting organism outcompetes the proving organism. The capacity for proof is a *luxury feature*—useful in safe environments with ample time, but not the primary mode of operation.

The two modes:

- **Trust** (*fides*) is the *base mode*—always running, metabolically cheap, evolutionarily ancient
- **Rationality** is the *rare mode*—occasionally activated, metabolically expensive, evolutionarily recent

We do not switch from rationality to trust when we are tired or stressed. We switch from trust to rationality when conditions permit—and then switch back as soon as they don't.

The illusion: We tell ourselves that rationality is primary and trust is a fallback. The truth is the reverse. Trust is primary; rationality is the occasional overlay. We are trusting beings who sometimes reason, not reasoning beings who sometimes trust.

This explains why:

- Arguments rarely change deeply held beliefs (the belief was not produced by argument)

- People defend absurd positions with apparent logic (reason serves belief, not the reverse)
- "Rational" people disagree on fundamental questions (their rationality operates on different trust foundations)
- Education in logic does not make people less biased (bias is not a failure of logic but a feature of trust)

Homo fidens does not reason its way to a worldview. *Homo fidens* inherits, absorbs, and constructs a worldview through trust—and then uses reason to defend, refine, and articulate what was already believed.

A.9 Corollary 9: The Bacterium as Rational Ideal

If rationality means "absence of unjustified assumptions," then the bacterium is more rational than the human.

This is not a joke. It is a direct consequence of the Complexity-Credence Trade-off.

The bacterium as pure verifier:

- Its world is small enough that its verification bandwidth covers nearly everything relevant to its survival
- It does not construct hypotheses about the future; it does not trust traditions; it simply reacts to chemical signals *here and now*
- It operates with minimal *fides* and maximal direct verification
- In this sense, the bacterium is the *ideal of the evidential approach*

The human as rationalizer, not rational:

- Humans *appear* rational to themselves only because they are skilled at self-deception
- We construct elaborate narratives to conceal the fact that all our actions rest on blind trust in memory, logic, and the stability of the world
- Our "rationality" is often merely an advocate writing a defense speech for decisions already made on the basis of *fides*
- While animals (and bacteria) are "cognitively honest" in their trust, humans expend enormous resources creating the *illusion* of proof

Why the bacterium cannot be us:

The paradox is that the bacterium's "rationality" is a consequence of its simplicity.

- As soon as a system wants to become complex—to have long-term plans, large body size, rich behavioral repertoire—it *must* begin to trust
- Trust is the *price* of escaping the prison of immediate chemical reaction
- If we tried to be as "rational" (verifying) as the bacterium, we could not take a single step without first checking whether the floor will hold

The inversion of hierarchy:

Traditional view:

$$\text{Bacterium (primitive)} \rightarrow \text{Animal} \rightarrow \text{Human (rational apex)}$$

ETC view (by verification capacity):

Bacterium (near-complete verification) $\rightarrow$ Animal $\rightarrow$ Human (minimal verification, maximal *fides*)

The bacterium is indeed closer to the ideal of "evidential cognition"—but this ideal is an *evolutionary dead end*. We became *Homo fidens* because it was the only way to become more complex than a bacterium.

The brutal conclusion: Rationality, as traditionally conceived, is not the crown of evolution. It is a *luxury feature* available only to simple systems with small decision spaces. Complexity requires trust. The more complex the system, the less rational (in the verificational sense) it can afford to be.

Summary: These corollaries establish that **Homo fidens** is not a description of human limitation but of human (and all complex system) *architecture*. We do not trust because we fail to know. We trust because trust is the only mode of cognitive operation available to finite systems facing infinite complexity. Knowledge, proof, and verification are valuable achievements—but they are achievements *within* a trust-based system, not escapes from it.

The animal, lacking the apparatus of rationalization, reveals what we truly are: trusting beings who sometimes verify, not rational beings who occasionally trust. *Homo sapiens* is the name we gave ourselves. *Homo fidens* is what we are.